

Peltier modules with compact footprints

CUI Devices has expanded its Peltier module product line with the addition of micro Peltier modules. The product's press release promises reliable solid-state construction, precise temperature control, and quiet operation for space-constrained designs where forced air cooling is not an option.



The possible applications of these devices include medical, industrial, refrigeration, and sealed environments. The micro Peltier modules are now available in compact package sizes with dimensions from 3.4mm to 9.5mm and profiles as low as 1.95mm.

CUI Devices

www.cuidevices.com

DC-DC converter

BMR683, a new series of digital PMBus DC-DC converters, has been launched by Flex Power Modules. The new series provides indispensable monitoring functionality and control of the feedback loop to help stabilisation, claims the company.

The company states that the high-density BMR683 saves valuable board space and has an output power of up to 500W and an input voltage range of 36V to 60V. Delivering an excellent price/performance ratio, it has a high efficiency at 95.5 per cent at full load and a broad output voltage adjustment range of 14V to 35V, allowing for easy optimisation of the connected RF section.

As per the company's press release, the device also allows users to configure several aspects of the device operation, including voltage and current for improved fault tolerance and optimisation of the module's control loop for stable operation in a wide range of circumstances.

Flex Power Modules

www.flexpowermodules.com

MOSFET

Recent technology trends for switched-mode power supply (SMPS) design in industrial applications include the need for high efficiency and power density as well as increased bus voltages, calling for the need for power devices with 650V breakdown voltage.



Infineon Technologies, with its 650V CoolMOS CFD7 product family, helps meet these demands. The company claims that these devices are best suited for resonant topologies in soft-switching applications like telecom, server, solar, and off-board EV charging.

For SMPS applications, these MOSFETs provide outstanding light-load and improved full-load efficiency. Furthermore, best-in-class RDS (on) enables customers to increase the power density level of SMPS at a competitive price.

Infineon Technologies

www.infineon.com

Electromagnetic interference capacitor

The new R52 series of compact polypropylene film X2 EMI (electromagnetic interference) suppression capacitors has been launched by KEMET.

The series fulfils the growing need for automotive, industrial, consumer and energy applications that require a smaller, high-capacitance X2 class solution for suppressing EMI. As per the product's press release these capacitors possess highly reliable standards required for smart grid hardware located in limited access areas where regular service is not available.



The R52 series can be used in DC-DC converters within on-board battery charger systems found in electric and hybrid vehicles.

KEMET

www.kemet.com

Power-saving microcontrollers

System developers of battery-powered and other power-conscious designs with and without LCDs can now easily add an array of innovative features using

a new line of PIC eXtreme Low Power (XLP) microcontrollers (MCUs) from Microchip Technology.

Packed with fourteen types of core independent peripherals (CIPs) that operate outside the central processing unit (CPU) for power savings, the PIC24F GU and GL families of



MCUs feature the CIP called LCD with autonomous animation. Capable of operating in harsh environments, the MCU families are ideal for automotive, industrial, and consumer applications.

The company states that the new MCU families are designed to increase the security of an application, whether it is connected to the Internet or a standalone system.

Microchip Technology

www.microchip.com

Image sensor

ON Semiconductor has introduced a high-performance AR0234CS 2.3MP CMOS image sensor with global shutter technology,



which has been designed for a variety of applications, including machine vision cameras, AR/VR/MR headsets, autonomous mobile robots (AMRs), and barcode readers.

The AR0234CS captures 1080p video and single frames, operating up to 120 frames per second (fps). With great shutter efficiency, the 2.3MP sensor produces crisp and clear images by minimising frame-to-frame distortion in high-speed scenes.

The AR0234CS sensor's innovative pixel architecture delivers the high dynamic range needed for supporting a variety of lighting conditions—from night to bright sunlight.

ON Semiconductor

www.onsemi.com

Crystal oscillator

A small SMD oven-controlled crystal oscillator (OCXO) has been launched by Raltron for better temperature stability in applications such as 5G wireless